Social Benefits Settlement Module

1. Introduction

The Social Benefits Settlement module is an essential part of the system, as it allows for the calculation and management of the labor benefits that must be granted to employees under Colombian law. This module ensures proper handling of severance pay, interest on severance, service bonuses, and accrued vacation days, optimizing compliance with labor obligations.

2. Objectives

- Automate the calculation of social benefits.
- Ensure that settlements comply with Colombian labor legislation.
- Generate a history of social benefit settlements.
- Allow consultation and export of benefit reports.

3. Features

3.1 Severance Calculation

- One month’s salary for each year worked.
- Proportional amount if the employee has worked less than a year.

3.2 Interest on Severance Calculation

- 12% annual interest on the total severance accumulated.

3.3 Service Bonus Calculation

- Paid in two installments (June and December).
- Equivalent to half a month’s salary for each semester worked.

3.4 Vacation Calculation

- Accrued according to time worked.
- Proportional days if the employee has not completed a full year.

4. Database Model

The following table(s) are required for this module:

Table: SocialBenefits

Campo	Tipo de Dato	Descripción
id	INT (PK)	Identificador único
empleado_id	INT (FK)	Referencia al empleado
cesantias	DECIMAL(10,2)	Monto de cesantías acumuladas
intereses_cesantias	DECIMAL(10,2)	Valor de intereses sobre cesantías
prima_servicios	DECIMAL(10,2)	Valor de la prima de servicios
vacaciones	DECIMAL(10,2)	Días de vacaciones acumuladas
fecha_liquidacion	DATE	Fecha de cálculo

5. Processes and Algorithms

Severance Calculation

- Retrieve the employee's base salary.
- Multiply by the number of years worked or the corresponding fraction.
- Store the result in the database.

Interest on Severance Calculation

- Take the total severance accumulated.
- Apply the 12% annual rate.
- Store the result in the database.

Service Bonus Calculation

- Determine the number of months worked in the semester.
- Apply the formula: (Base salary / 12) * Months worked.
- Store the result in the database.

Vacation Calculation

- Verify the total time worked.

- Apply the formula: (Days worked * 15) / 360.
- Store the result in the database.

6. Conclusion

The Social Benefits Settlement module enables efficient management of employee benefits, ensuring legal compliance and preventing calculation errors. Its integration with other system modules ensures proper handling of employee information and their earned rights.